Bike Frame Design Project

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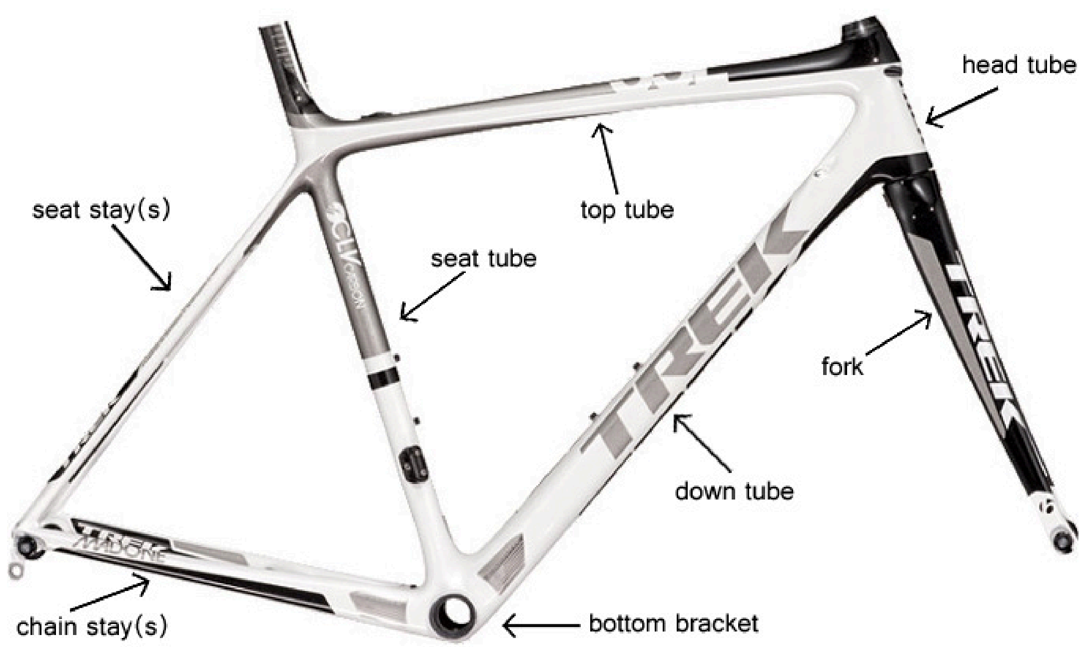
MAE 3150

Introduction:

This project targets the design and analysis of a bicycle frame used for an electrical bike share program. The importance of designing a frame suitable for such programs is accessibility and affordability. Bikes in these programs would allow for easier travel around communities, including college campuses such as Cornell. This would give students access to a reliable transportation method while decreasing the dependence on cars and public transportation. Other than just transportation, the usage of bicycles promotes physical activity and contributes to better health.

The goal of the analysis is to assess desirable characteristics, such as being lightweight, durable, inexpensive, and accessible. Through the use of Fusion360 and ANSYS, two frame designs were analyzed with realistic loads to assess deformation, stress, and the factor of safety throughout the structure. The desired factor of safety threshold is at least 2.

Terminology:
Below are component names of the bike chassis that I will be using in this document.

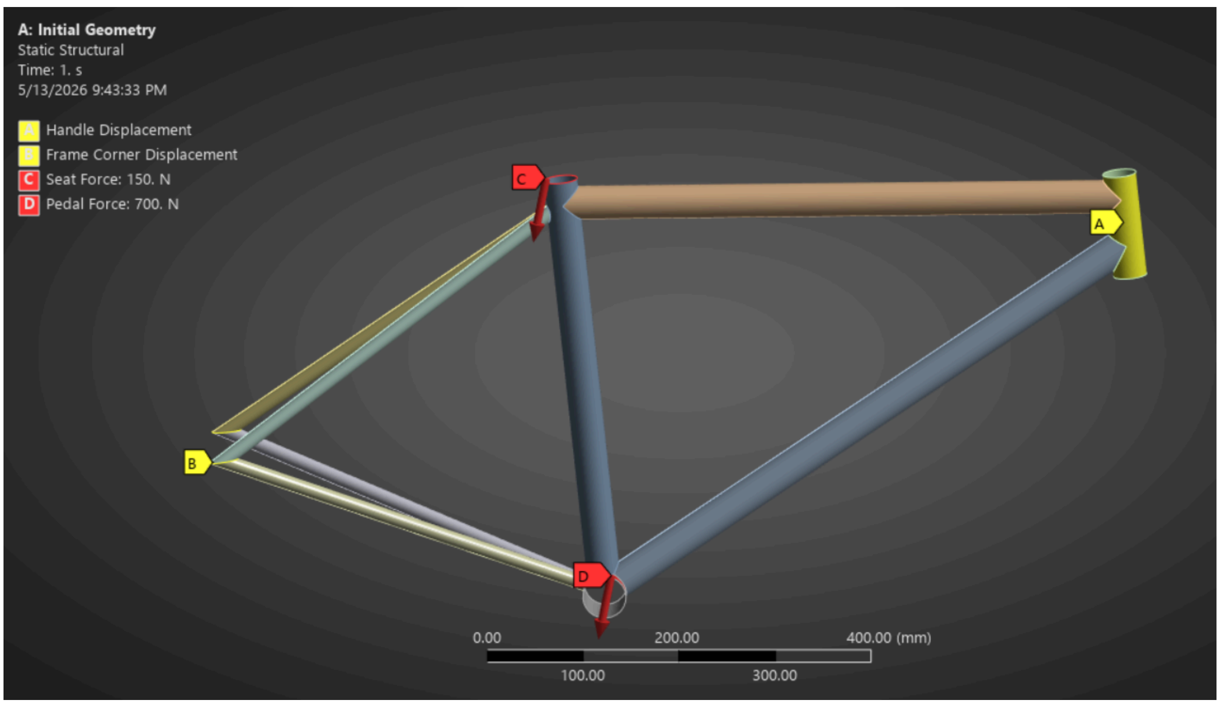


Source: https://www.shoptrekbikes.com/about/basic-bicycle-anatomy-101-frame-pg433.htm

<u>Analysis of the Provided Geometry</u>:

The bike frame geometry was created using Fusion 360, consisting of surfaces that represent the components of a bike chassis. Using surfaces allows the configuration of the thickness on ANSYS. The initial configuration of the bike frame contains seat, top, down, and head tubes of diameter 35 mm. The chain and seat stays have a diameter of 15 mm. Exporting the CAD model from Fusion 360 to ANSYS, the structure was then analyzed using shell theory with SHELL181 elements. Shell theory was used to generate 6 degrees of freedom at the nodes of the curved elements, representing x, y, and z displacements and rotations. The chosen material for the models is aluminum alloy. The initial model used a thickness of 2 mm and an element size of 4 mm.

Below are the loading conditions for this structure:

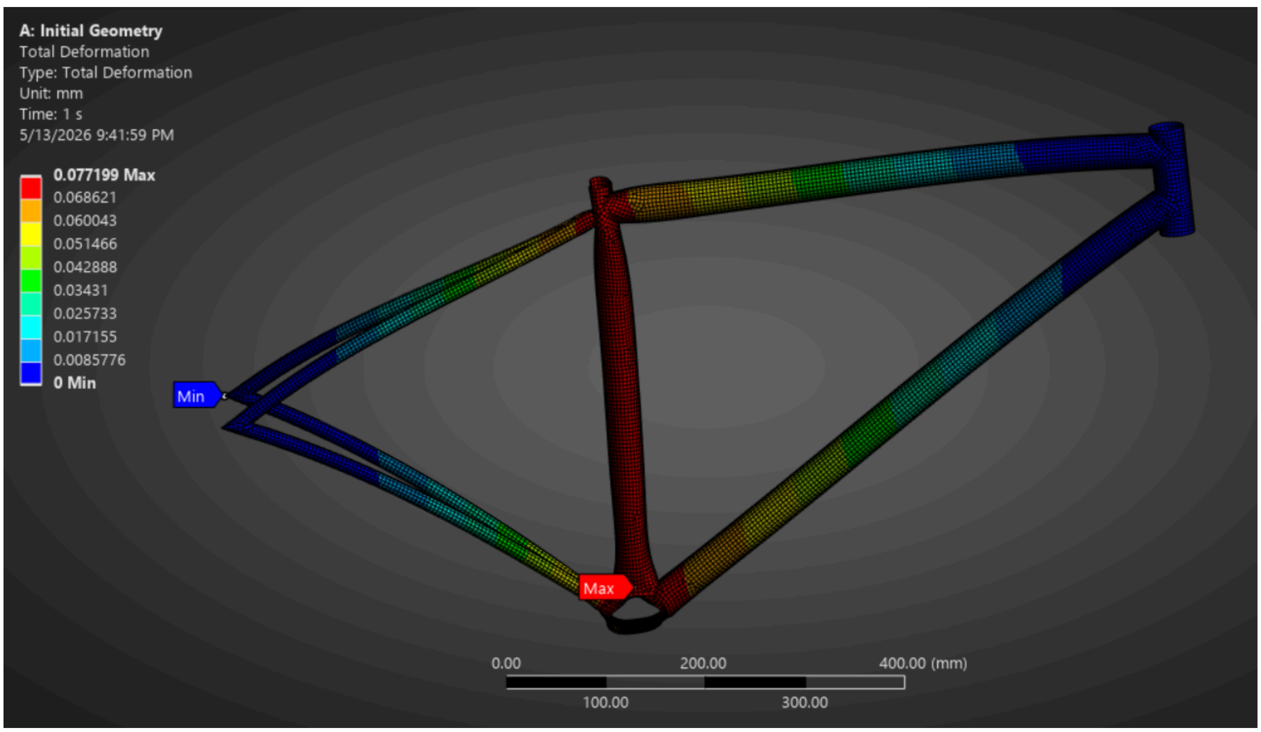


[A] and [C] represent fixed supports for the head tube and the corners for the rear tubes respectively. [C] and [D] represent loads that would appear when a person is using the bicycle – the user's weight and the force they exert on the pedals.

The above boundary conditions and geometry were used by ANSYS to evaluate the deformation, equivalent stress, and factory of safety contours of the structure. These are assessed to determine whether the geometry is structurally safe and meets the goals of this project. ANSYS evaluates these properties by creating a mesh using the aforementioned element size of 4 mm and creating algebraic equations derived from the strain energy and external work done on regions adjacent to the nodes.

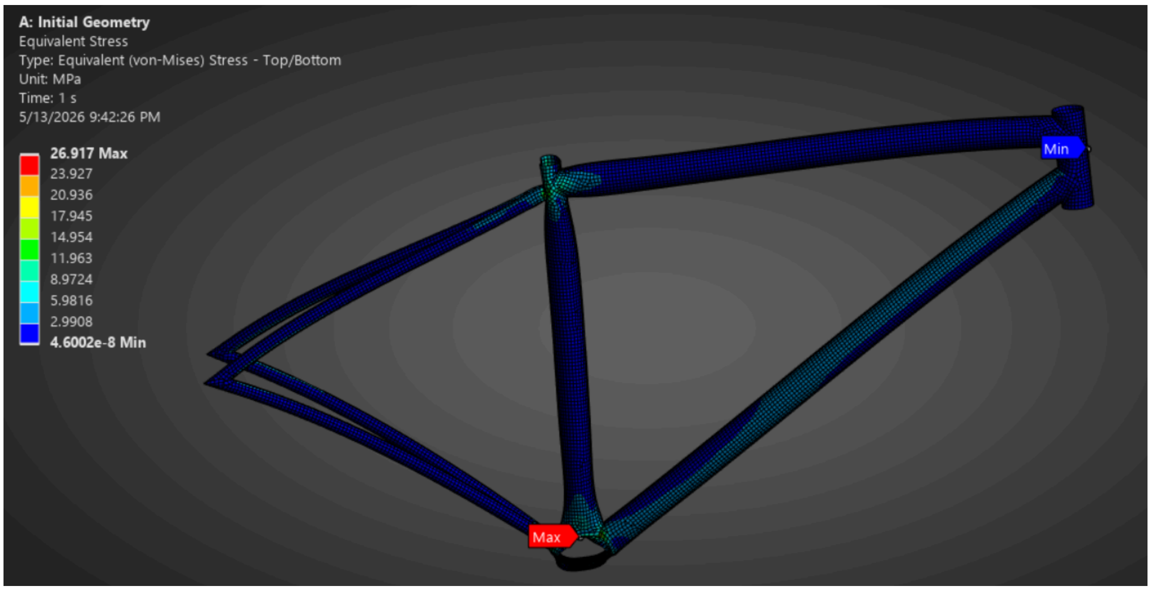
Total Deformation:

The simulation results for the total deformation shows that the greatest deformation under the loading conditions is about 0.0772 mm, whereas the minimum deformation is 0. The maximum deformation occurs along the seat tube, the bottom bracket, and spreading to adjacent components. The deformation gradually decreases along the tubes further away from the seat tube and bottom bracket. The deformation contour is very reasonable, as the head tube and the rear corners are fixed (no rotation or translation). The seat tube and bottom bracket having the greatest deformation makes sense due to the seat and pedal forces being applied on those structures respectively.



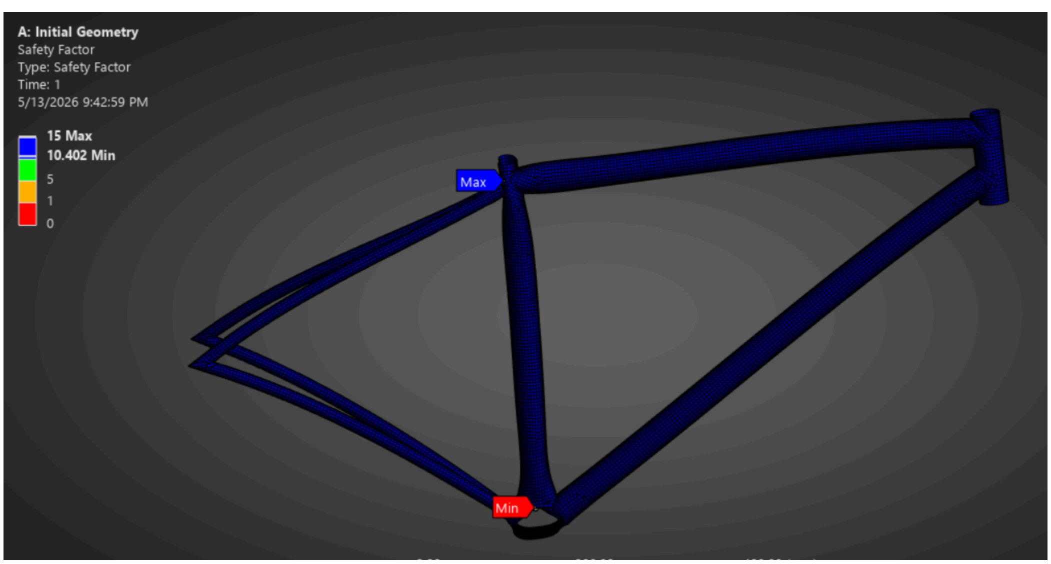
Equivalent (Von-Mises) Stress:

The stress contour shows that the maximum equivalent stress occurs on the bottom bracket with a value of 26.917 MPa. The minimum stress is approximately 0, with the majority of the structure having very low stress. The high stress value on the bottom bracket likely comes from the adjacent rods connecting to the component, as well as the high pedal load.



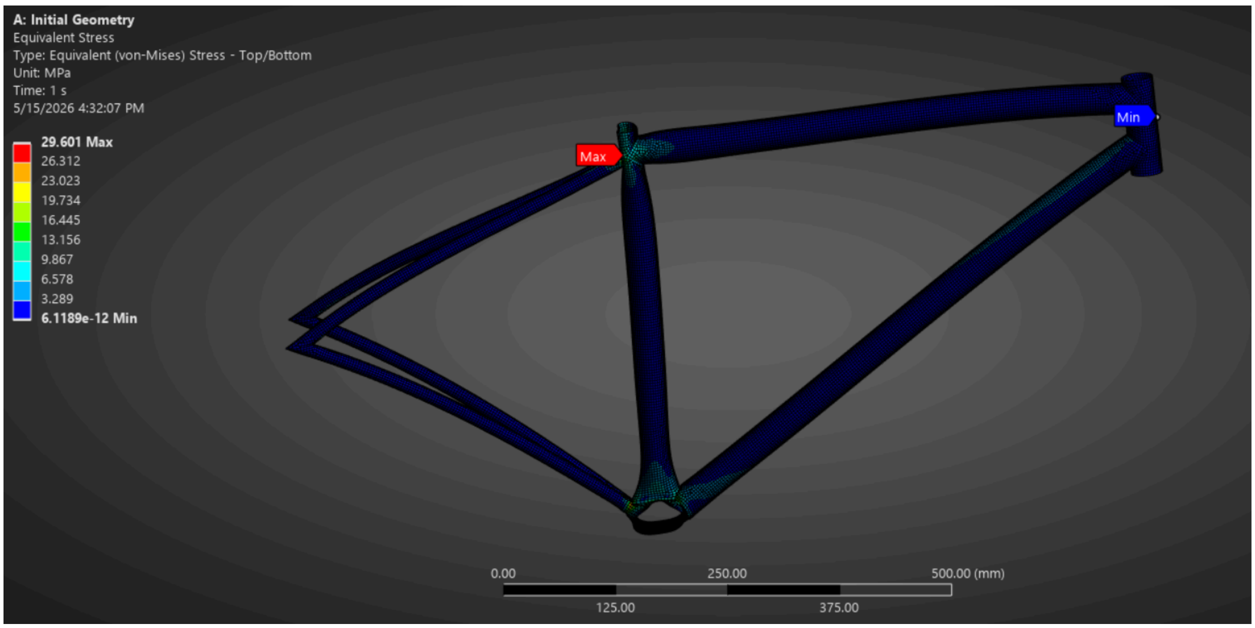
Factor of Safety:

The minimum factor of safety of the overall structure is 10.402, which passes the goal of achieving a value of at least 2 in every component.



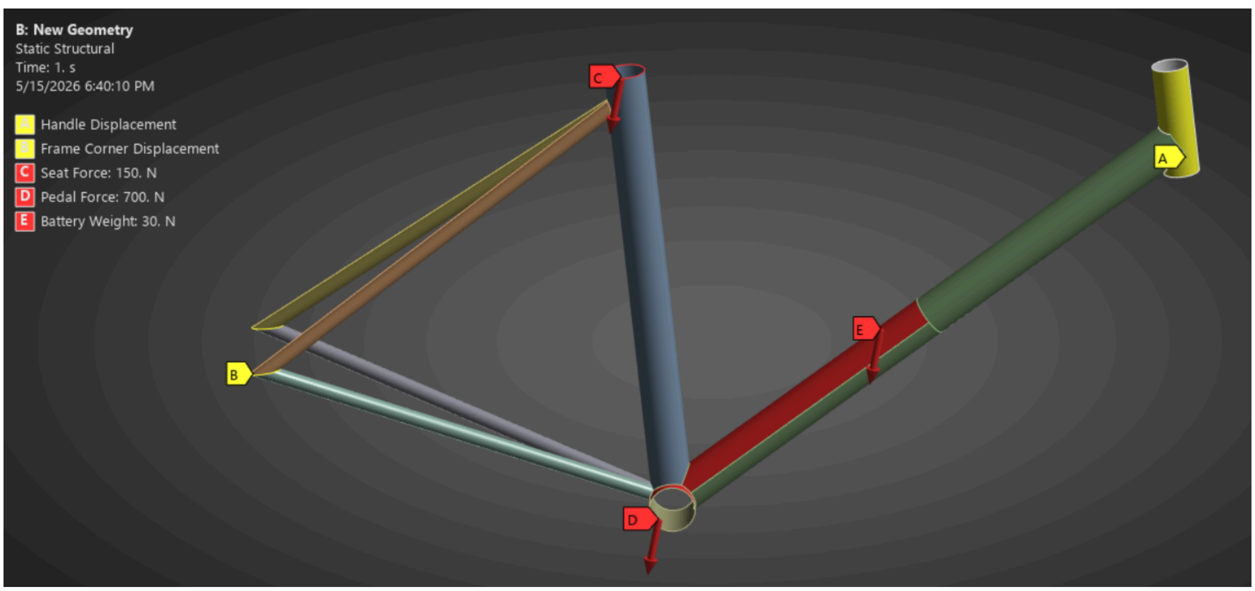
Simulation Verification:

To verify the results found previously, I reduced the element size for the mesh from 4 mm to 3 mm. This causes there to be more elements in the mesh, making measurements more precise. The deformation and factor of safety values were identical as the solution with the 4 mm elements, whereas the stress values slightly increased. This comes with the increased precision of having more elements. Because not much has changed except for a small increase in stress, the previous results are reasonable.



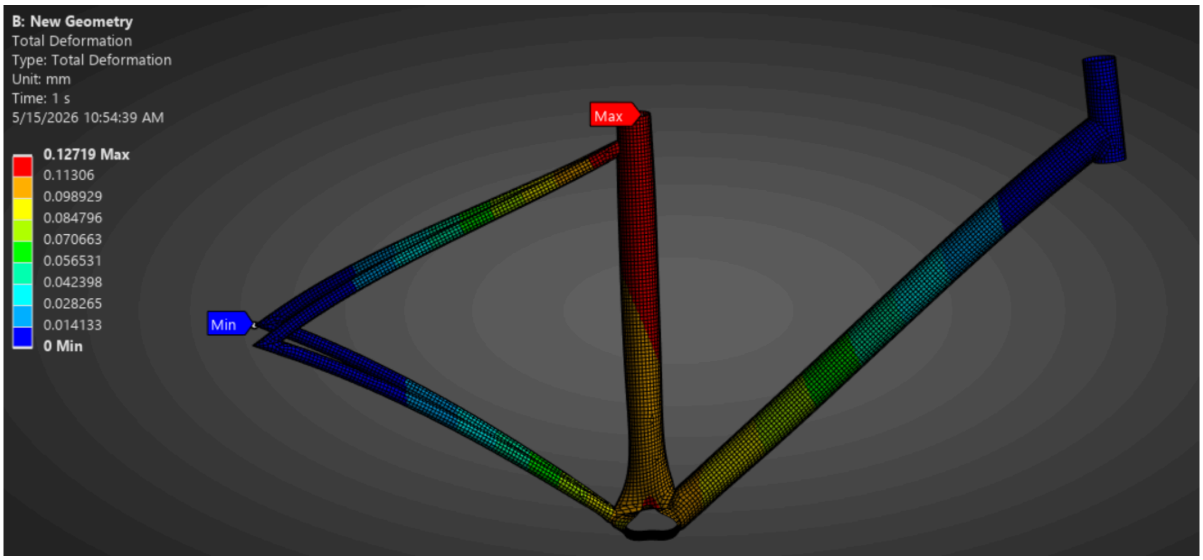
Analysis of the New Design Geometry:

To promote accessibility in a bikeshare program, it can be considered for a bike frame to have its top tube removed. This allows users to easily position themselves to sit on the bicycle by reducing the height needed to get one leg over the bicycle. Another consideration I made for this geometry was to bring the head tube of the bicycle closer to the seat. This was done to reduce the moment arm induced by the user holding the handle. To do this, the distance between the head and seat tubes was reduced from 582.734 mm to 553.372 mm. I also changed the diameters of the seat tube and the down tube from 35 mm to 40 mm to increase the moment of inertia for which the loads are applied. I expect this change to decrease the stresses about the bottom bracket. The rest of the geometry remained unchanged as well as the material of the frame (aluminum alloy). The loading conditions for this design are different however, as there is an additional battery load along 300 mm of the down tube. This was done to simulate the battery used for the electric bike since it has its own weight. The other boundary conditions remain the same.



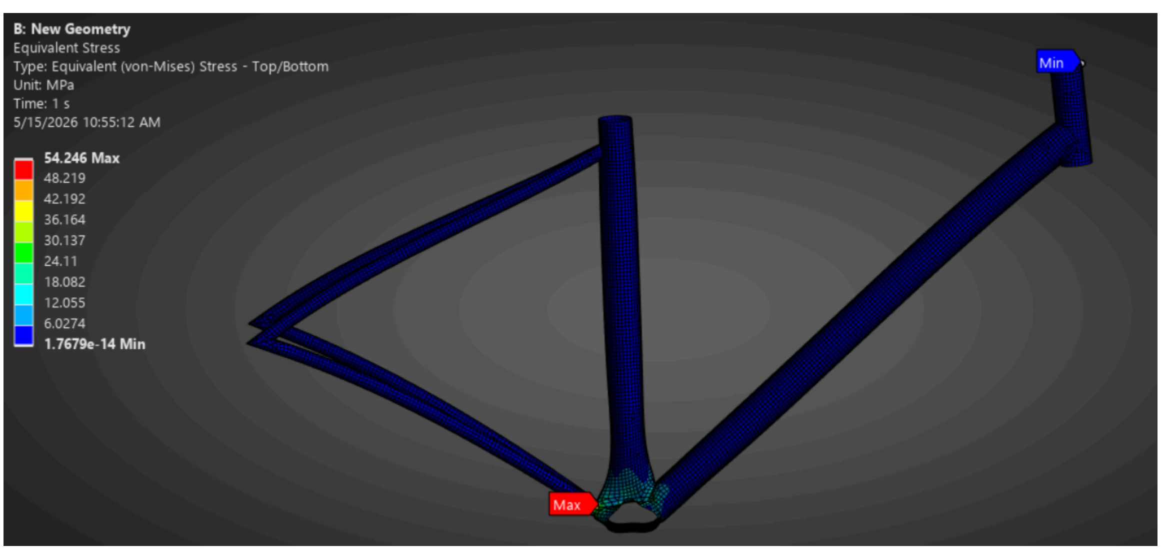
Total Deformation:

The maximum deformation of 0.1272 mm occurs closest to where the seat load is applied and partly along the seat tube. The minimum deformation occurs at the essential boundary conditions just as the initial geometry.



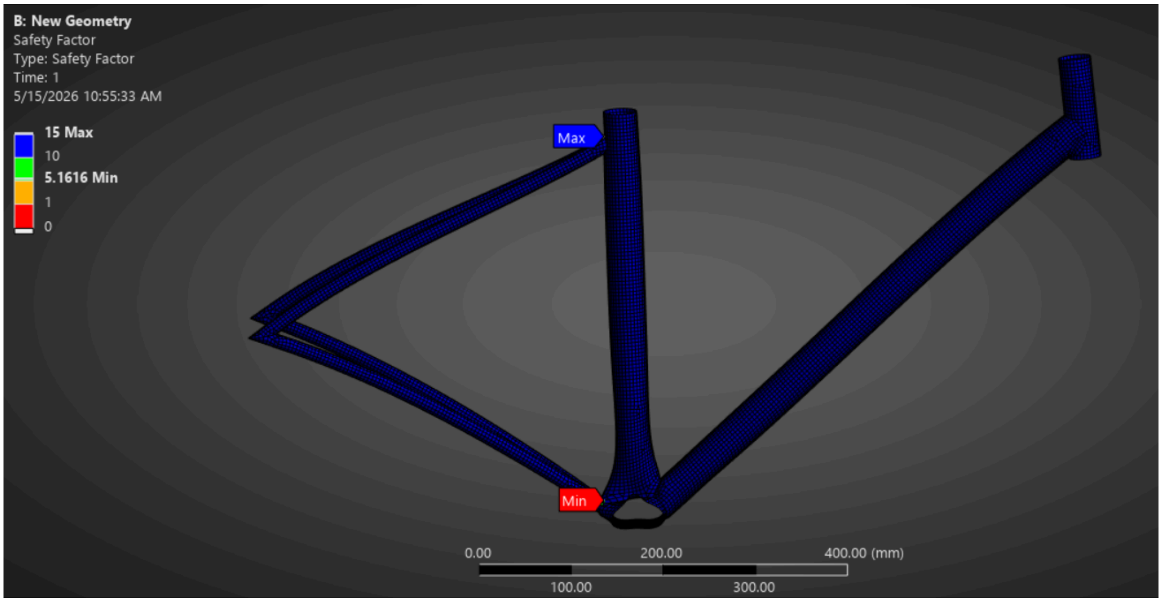
Equivalent (Von-Mises) Stress:

From the equivalent stress contour, the maximum stress occurs at the bottom bracket just as in the initial geometry, however, the stress is much greater at a value of 54.246 MPa. This is reasonable since the battery weight acts in the same direction as the pedal and seat loads.



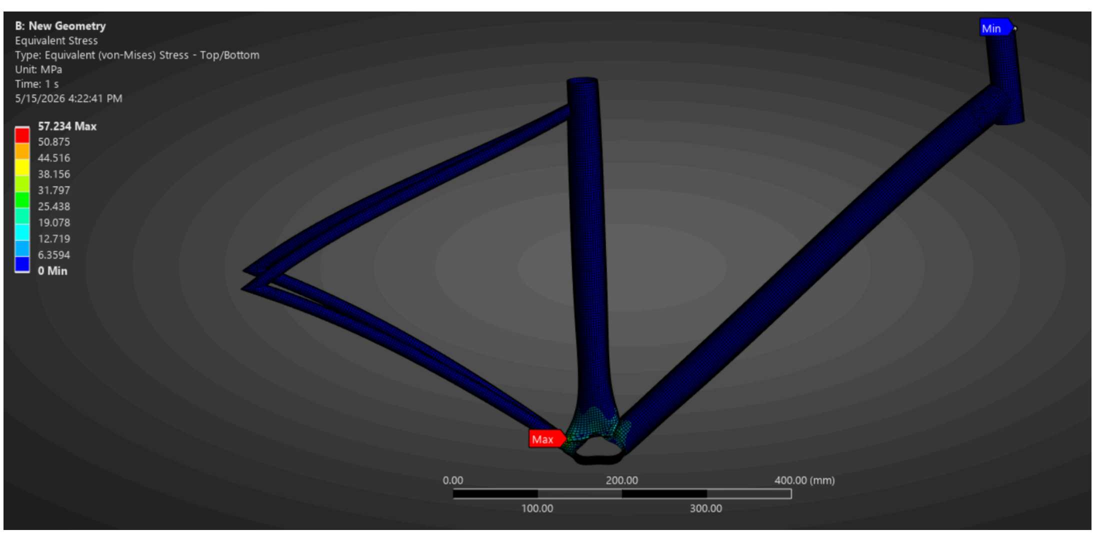
Factor of Safety:

Based on the results, the minimum factor of safety of the structure is 5.162, which also passes the goal of having a factor of safety of at least 2.



Simulation Verification:

As done with the initial geometry, element sizing was changed to 3 mm. This refined the mesh and provided the same deformation and safety factor contours as the 4 mm mesh. Similar to the previous geometry, the stress also slightly increased, however that is expected with a more refined mesh. Therefore this model is reasonable.



Optimization:

In order to optimize the bicycle frame design, it is important to reduce the mass of the frame while maintaining a factor of safety greater than or equal to 2. Reducing the mass connects to the goal of making the e-bikes accessible and inexpensive since reducing the mass means there's less material used. In the optimization process, the thickness of the seat and down tubes were chosen as the primary design parameters since I altered their diameters for the new design geometry. The other components had their diameters unchanged at 2 mm. The allowable range chosen for the parameterized thicknesses was 2 mm - 4 mm. Candidate points were chosen with pairs of thickness that are equal. This means that the candidate points had the seat and down tubes having a thickness of 2 mm, 3mm, and 4 mm. The optimization results showed that the average mass of the seat tube was about 0.292 kg, which essentially didn't vary at all from the mass of the tube prior to the optimization. The down tube mass on the other hand had an average mass of 0.5541 kg. The lowest factor of safety from the optimization turned out to be 5.1625, which is still well above the target value of 2. This means that the model has a very good mechanical advantage and is safe.

Table of Schematic D4: Optimization , Candidate Points

	A	B	C	D	E	F	G	H	I
1	Reference	Name	P1 - Body39 Thickness (mm)	P2 - Body40 Thickness (mm)	P9 - Body39 Mass (kg)		P10 - Body40 Mass (kg)	P17 - Safety Factor Minimum	
2					Parameter Value	Variation from Reference		Parameter Value	Variation from Reference
3	◉	Candidate Point 1	2.0004	2.8098	0.29214	0.00%	0.58312	5.3029	0.00%
4	○	Candidate Point 2	2.0004	3.161	0.29214	0.00%	0.6623	5.361	1.10%
5	○	Candidate Point 3	2.0005	2.0005	0.29215	0.01%	0.41691	5.1625	-2.65%
*		New Custom Candidate Point	3	3					

<u>Reflection</u>:

This project is a culmination of the skills and knowledge I learned from taking MAE 3150. Taking part in making a design and simulating it brought about a deeper understanding of the workflow that engineers go through to design products. It always is fascinating to see the colored contour of a structure and see how the overall shape changes due to its environment.

The most difficult part of this project was finding changes to cover for the new design. There were iterations where I wanted to test different angles, however after importing the CAD file into ANSYS, there were issues with meshing and I had to scrap the idea. This in turn made me spend much more time redesigning the CAD. In the future, I certainly will practice using the software more and be able to simulate things I want to while understanding issues that may arise like what I encountered. Overall, it is apparent how much I learned from this course, and I can't wait to simulate more designs.

Appendix:

The ANSYS crosstalk, “Thinking Like an Analyst”, delves into the process of engineering analysis using ANSYS. Before setting up a simulation, the engineer must define the objective of the analysis. This determines the geometry to include, the materials to define, and the analysis type to use. The general workflow for building a model is to define the objective, select an analysis type, simplify the geometry, idealize the mechanical system, apply boundary conditions, and assign material models.

The objective is the first step to analysis, which essentially asks what is the problem that you are trying to solve. The analysis type can be chosen among a variety of options. For a bike frame under static loading, a static structural analysis is appropriate. Dynamic components like suspension systems require transient or explicit dynamics instead.

After selecting an analysis type, there is geometric simplification. Removing irrelevant features such as logos and small fillets reduces computational cost and time taken for an analysis. For thin-walled tubular structures like bike frames, shell elements are generally preferred over solid elements. Using solid elements would produce an unnecessarily fine mesh through the tube wall thickness. Connections between parts can be modeled at different levels of fidelity. Contact regions are the most detailed, modeling surface interactions point-by-point. Joints provide an idealized representation of part kinematics with fewer degrees of freedom. Springs and beam connections offer an intermediate approximation of fastener stiffness. As an example, bolts can be idealized by suppressing the bolt body and inserting a beam connection between the relevant faces.

The guiding principle for boundary conditions is to consider what is not modeled at each boundary and how that missing component would interact with the model. Every part removed from the model involves a tradeoff where the engineer must think carefully about what physical behavior is lost. For distributed loads such as battery or seat weight, a point mass promoted to a remote point allows ANSYS to compute the resulting forces on the structure.

Whether to include nonlinear material behavior depends on the objective. For an aluminum frame designed to remain below yield, a linear elastic material model is sufficient. If the simulation produces stresses above the yield strength, this already indicates a failed design without needing a plasticity model. Nonlinear models add complexity and should only be used when the objective specifically requires capturing post-yield behavior.

The crosstalk discusses the aspects of problem solving that are essential for analysis using ANSYS. The software is merely a tool, and the results can vary depending on the approach taken.